

H O B B S S

NEW DISCOVERY

FOR LONGITUDE



ROYAL OBSERVATORY

H O B B S S

NEW DISCOVERY

FOR LONGITUDE



ROYAL OBSERVATORY

The Project Gutenberg eBook of A new discovery for finding the longitude

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: A new discovery for finding the longitude

Author: William Hobbs

Release date: January 4, 2025 [eBook #75039]

Language: English

Original publication: London: William Hobbs, 1714

Other information and formats: www.gutenberg.org/ebooks/75039

Credits: John Welch, MWS, Cambridge Digital Library and the Online Distributed Proofreading Team at <https://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK A NEW
DISCOVERY FOR FINDING THE LONGITUDE ***

A
New DISCOVERY

For Finding the
LONGITUDE.

Humbly Submitted to the

Approbation of the Right Honourable the Lords Spiritual and Temporal,
and the Rest of the Honourable Persons, appointed by the late *Act of
Parliament*, for Hearing and Determining *Proposals* relating to the
said LONGITUDE.

By William Hobbs, *Philo Mathem.*



LONDON:

Printed for the Author, and are to be Sold by *Fard. Burleigh* in *Amen-
Corner*.

Price Six-pence.

Note, The former Impression was Printed *July 12. 1714.*

A New Discovery for Finding the Longitude.

For Finding the Longitude, it is to be noted, That if the true Hour, and Minute of the Day, at the Place of our first Departure, and where we are Arrived, can be both obtained, the Difference of Longitude may be found, as truly as the Difference of Latitude.

For which purpose the *Spring-Movement* hereunto subjoined, is most humbly proposed to effect the same, even to the Tenth of a Minute of Time. So that should the Error of a Minute happen to be found in Practice, it must proceed from the Influence or Vicissitude of the Air, and not from any Defect in the Rules by which the said Movement is composed, as will plainly appear by the Explication thereof, in the following Lines; and first of

The Description of the said Movement.

This Movement may be either of the Form and Bigness of that in a Common Clock; or else of the Smallness of a Pocket Watch, as may be thought convenient in Practice. But the larger it is, the better the Divisions in the *Horologe* will appear. And as for the Numbers, they may be at pleasure: Only this must be observ'd, That the swiftest Index must make just One Revolution, whilst the slow one goes but One of the Hundred Divisions, (as is common in Minute Watches, wherein 60 Minutes goes to an Hour) but the rest of the Numbers may be at pleasure, as aforesaid. And if the Slowest moves round in 3, 4, or 5 Days; and the Movement requires to be wound up once in 26 or 27 Hours, it will be sufficient: For the more Vigor the Motion is calculated for, the less the Air will affect it.

How to set it first in Motion.

When it is finished, let it go for a Month or two; after which put the two *Indexes* right one with the other, and both of them pointing just to 100. Then the Spring being wound up, keep it from Going (with the Key) till the Shadow of the Sun is exactly come to a *Meridian* Line, which must be rightly prepared for that purpose. Then let it go till the Shadow comes just to the same Line the next Day, or rather 40, 50, or 100 Days after, (the longer the Better.) At which time enter down the *Centesms* that the swiftest *Index* points to, as also the *Integers* and *Tenths* of the slowest, (but put the last first in your Numbers) by this you will be furnished with two general Numbers, viz. the said Numbers pointed to, and the Time (in Hours) spent therein, to find the Hour of the Day to the Tenth of a Minute, at any time; for that Place where it was so set going, tho' you remove it afterwards to any Distance whatsoever. Which two Numbers ought to be entered in a Book; to be used whensoever you would find the Hour of the Day.

And it is to be noted, That if the Sun be not then in its Mean Motion, you must Add to, or Subtract from what the *Indexes* do Give, according to the Inequality thereof: But if you try it for one whole Year, there will be no need of either. And that what is said may be the better understood, I shall give an Example.

How to find the Hour and Minute of the Day at London, by the said Movement.

Suppose it should be made for the Slowest to Revolve in about five Days, and after it has gone just two Days, the Slowest *Index* should then point between 39 and 40, and the Swiftest to 21.5. Then both these Numbers (as set in Order) will make 3921.5. And if the Sun be in its Mean Motion (if not, you must Add or Subtract, as aforesaid) then the said Numbers 3921.5 must always be the First; the said two Days, or 48 Hours, the Second; and the two Numbers pointed to by the *Indexes*, at the time for which you would find the Hour and Minute of the Day (as suppose the slow one should point to 87, and the swift one to 65.2, both making 8765.2) must be the Third; by which the Fourth will be obtained as followeth: Which Fourth Number being Divided by 12, the Remainder will be the Hour and Minute of the Day at *London*, as was required.

	Hours		Hours.	
As	3921.5	:	48	:: 8765.2 : 107.18
				ho. min. }
			Remains 11	: 10.8 } Requir'd
			hours.	}
			or 11 ¹⁸ / ₁₀₀	}

And so for any quantity of time less than one Revolution.

But if the time required to find the Hour and Minute at *London*, (when at Sea or Land,) be some Weeks or Months, *after* it was first set going; or (if cleansed) after it was last *Cleansed*; then you must add 10000.0 to what the two *Indexes* do shew, for each *Revolution*, that it has made, since it was so set going, or last cleansed. And if you would at any time know how many such *Revolutions* must be added, to what is shewn by the Horologe; you must first note the *Day* of the Month, that you are then seeking the Hour for; as suppose *Novem. 12. 1715*. After which note the *Day* last entred, when cleansed; as suppose *Sept. 5. 1715*. Then compute the number of Days between *Sept. 5.* and *Novem. 12.* which is 25, 31, and 12, In all 68. After which observe the *Time* last spent in one Revolution; which you see was 5 Days, 4 Hours, and $\frac{1}{8}$ (for which set down $\frac{1}{10}$ for 'twill make no sensible Difference) and having thus done, say,

	Days	Hours		Hours	Revo.		Days
As	5	4 $\frac{1}{8}$	equal to	124.1	:	1	:: 68 : 13. whole Revolutions;

For which, as before directed, you must add 13 times 10000.0 to the Numbers pointed to, by the two Indexes: Which if we suppose 3752.2 making in all 133752.2 will be the Number required, for the 12th of *November*, if the Indexes should both point as before proposed. And having now found this Number, you are prepared to find the Hour and Minute at *London*, in manner following (*viz.*) suppose you are in or near 44 Degrees of Latitude, in the Autumn Season. Then seek the Latitude 44 Deg. and the Season *Autumn*, in the Book that is peculiar to your Movement, and having found both, take the Numbers Answering thereunto, which (in this Example) is 29274.2 and 240 Hours; and then say,

	Hours		Hours
As	29274.2	:	240 :: 133752.2 : 1096.84

Which being devided by 12, leaves remaining 4.84 ho. or 4 ho. 50.4 min. required, for the Hour and Minute at *London*, by which the Longitude will be found, as before directed.

And seeing it was near the Winter Season: If you should add the Autumn and Winter Numbers, and Hours respectively together; and make them the two first Numbers in your Proportion; it would *Æquate* the Seasons more exactly: I mean, if the said Numbers had been really entred from Practice; but these are only supposed. And by this method you may *Æquate* the Numbers for finding the true Hour and Minute at any Season; or for any other Latitudes whatsoever.

How to find the Hour and Minute, at the place where you are, when abroad, either at Sea or Land.

For Solving this, there are Two Ways, the First and Infallible is, when the Sun is either Rising or Setting; at which times, if the Air be clear, observe what Numbers are pointed to by the two *Indexes*, (but observe the swiftest first, because it will soon be removed) just as the Center of the Sun is in the Horizon; and enter both the Numbers down, as before directed, with the Day of the Month, and the Latitude you are then in: Which having done, by the following Cannon, say,

As P_x

Is to t. c. of the Sun's Decl.

So is t. Lat.

To s. c. of an Angle; which, if converted into Time (by 15) Gives the Hour and Minute of the Sun's Rising from Midnight; and if Subtracted from 12 ho. gives the Setting. Which shews the true Hour and Minute at the Place where you then are, either at Sea or Land.

The Second way for Solving the said Question, is by taking the Sun's Altitude, when about 5 or 10 degrees above the Horizon; and at the same time let another Person see what Number the Swiftest Index points to, and after that the Slowest: Then put them both down as before directed, with the Day of the Month, and the Latitude of the place where you are; then have you the Sun's *Altitude*, the said *Latitude*, and the Sun's *Declination*, to find the Hour and Minute of the Day at the place required.

Now having by the said Observations and Cannons found the Hour and Minute of the Day for the Place where you are, when abroad; and by the Numbers pointed to, by the two *Indexes*, also found the Hour and Minute that it is at *London*, (if the Movement was there set going;) If you take the Lesser of these from the Greater, and bring the Remainder into Minutes, it will give the difference between the two places in Time; by which you may find their difference of Longitude either *East* or *West* to a Minute, by the following proportion, (*viz.*)

Suppose the Difference of Time was 37 Minutes, then say,

$$\begin{array}{ccccccc}
 \{ \text{Minutes of } \} & \{ \text{Minutes of } \} & \{ \text{Minutes of } \} & \{ \text{Minutes of } \} \\
 \text{As } \{ \text{Time } \} & : \{ \text{Longitude } \} & :: \{ \text{Time } \} & : \{ \text{Longitude } \} \\
 \{ 1440. \} & \{ 21600 \} & \{ 37 \} & \{ 555.5 \} \\
 & \text{or 24} & & \text{Deg. Min.} \\
 & \text{hours} & & \\
 & & & \text{or 9 : 15.5} \\
 & & & \text{Required.}
 \end{array}$$

Which 9 deg. 15.5. min. being added to, or taken from the Longitude of the Place, where the said Movement was first set in Motion, gives the Longitude you are come into, as effectually as the Latitude is obtained by Observation, as was at first proposed. So that it is evident, that if any defect should be found in practice, it must proceed from the Vicissitude of the Air; and not from any Deficiency in this Discovery, or from the Cannons relating thereunto.

And that we may Answer all *Objections*, which may arise, on Account of the said Vicissitude of the Air, Motion of the Ship, Foulness of the Movement, &c. as well as render this Discovery more intelligible, we will descend to speak thereof more particularly: And,

In the first place, let Five or Six (or what Number you please) of these Movements be made; and after they have gone for a Month or two, (or if designed for the Sea, 'till within two or three Weeks of your Sailing) then take them asunder, and set them going by the *Meridian* Line, as before directed. And having entred down in the Book, herein after mentioned, the Day of the Month, when set going; how long it is making one Revolution; and what Number is produced, in just Ten Days Motion, (all which must be done, before you Sail in every Voyage;) Let them be carry'd to Sea, into any Latitude, (suppose 50 Deg. or *Newfoundland*) and at their Return bring them again to the said *Meridian* Line, and thereby (with the help of the Number of Days from their being first set in Motion) find the two first *Proportional Numbers*, as before directed: Which done, enter them in a Book, peculiar to each Movement, (the Form whereof followeth) together with the Latitude, and Season of the Year in which the Voyage was made. Then will they all be fitted for a Voyage to *Newfoundland*, and Parts adjacent, for any such Season. And so for any other, when Tried and Entred, as now directed.

After which, send them all in a Ship bound to the Latitude of about 37 Deg. (suppose *Virginia*); and being returned, let the Numbers be found and entred, as

aforesaid, with the Latitude and Season; then will any one, or all of them, be fitted for a Voyage, either to *Newfoundland* or *Virginia*, or any of those Parts, for any such like Season; and so for any other, when Tried and Entred as aforesaid.

This being done, send them also in a Ship, bound to the Latitude of 13 Deg. (suppose *Barbados* or *Jamaica*;) and being returned, find the Numbers, and enter them with the Latitude and Season: So will they all be fitted for these several Latitudes, (*viz.*) 50 Deg. 37 Deg. and 13 Deg. at such a Season: And by a like Process to the *Æquinocial*, and towards the *North Pole*.

Upon which several Tryals, you will have so many Movements fitted for all the *North Hemisphere*, both for Hot, Cold and Temperate Seasons. And, by only Counterchanging the Latitudes, for the whole Globe; or at least for the Marine Parts thereof.

And to what is proposed to be Entred, in respect of Latitude and Season, you may also add the Time of its being first set in Motion after Cleansing; which Cleansing must be done only Annually, and as near as may be on the same Day of the Month; which will much facilitate the Inequality arising from its Foulness: By which all Objections will be answer'd, and the Discovery render'd compleat and perfect. Especially if a true *Meridian Line* be made in every considerable Port in this Kingdom, and elsewhere; for Examining the Movement at your Arrival abroad, and setting it in *Motion*, before your Departure. But remember, that where you set it going, from thence your *Departure* must be reckon'd.

And by this Method all the said Vicissitudes that may arise in respect of Latitude or Seasons, Motion of the Ship (which will be near the same in one Voyage, as in another) and Cleanness or Foulness of the Work, will also plainly appear upon Record, (by comparing one Number in the said Book with another) and will be Accounted for, and *Æquated*, as occasion requires: Namely, by only taking such Numbers from the said Book, to work your Proportion by, as are entred for the Latitude and Season, that you are again Arrived to, or Sailing under: As suppose into 50 Deg. in the Summer Season, then take the Number and Hours answering thereunto; and so, if in Winter, those for the Winter. By which proportionable Numbers, the said Contingencies will be truly and effectually Allowed, and the Difference in Motions justly *Æquated*, as was before asserted.

Note, That in this Movement, the Time is Regulated or Conformed to the Motion: Whereas in former Movements, the Motion is Regulated or

Conformed to the Time. And this ought to be well understood, before you can rightly comprehend what is herein contained.

Now that this Movement, or Discovery, does not only differ from, but also far exceed, any that has hitherto been proposed; is evident in this (*viz.*) That if any former Movement whatsoever be Timed here, as usual, **Before** it is carried to Sea, (tho' never so exactly) yet the change of Climate, &c. will so affect it, as to make it go False: And if you fit it there, then 'twill be also False when brought hither: But if by this Movement, and the foregoing Rules, the said Numbers be found **After** it is returned from such a Voyage; all the said Defects or Contingencies will be *Æquated* by the said Numbers only. So that it will be the same in use, as if no such Accidents did attend it at all. And this Assertion must, and will be granted, by every one that does fully comprehend the Nature and Use thereof. For altho' this *Horologe* has no certain *Hours* or *Time* pre-assigned to it; yet it hath nevertheless, greater or lesser Numbers *Æquated* to the Variableness of Motion or Time, (*viz.*) If fitted to be used here in *England*, then it hath (or may have) one sort of Numbers (or Time) for finding the Hour and Minute, in the Summer Season; another for Winter; and a third for the Spring and Fall; one when it lyeth undisturbed in one and the same place; another (if you please) when carry'd always in a Rattling Coach. And on the other hand, if you carry it to Sea, then it has one for the *Torrid*, one for the *Frigid*, and a third for the *Temperate Zones*, &c. And yet still but one and the same *Horologe*, and but one and the same *Indexes*. So that all manner of Natural Vicissitudes or Contingencies whatsoever, may be Regulated or *Æquated* in and by the said Numbers; and consequently, the true Hour and Minute of the Day, where it is first set going, may be hereby as effectually found, as if no such Impediments did at any time or place attend it, as aforesaid.

And as for what may be suggested in relation to the Inequality between the Spring and Fusie, if any should be, it will be compensated, in that there are five or six Windings up of them, to one Revolution in the *Horologe*. So that there is nothing wanting to compleat this Discovery, but to put it into Practice: And seeing the Charge and Trouble is very Inconsiderable; and the Legislature by *Act of Parliament*, hath been graciously pleased to Incourage hereunto, it is thought expedient to make this publick Proposal, *viz.*

That if any one Person is willing to Engage in this so Noble an Enterprize, he will be readily received as a Partner, on such Conditions as may be Reasonably Concerted therein.

You may hear of the Author at Mr. Burleigh's in Amen-Corner,
London.

MEMORANDUM.

1. All Departures must be reckon'd from *London*, if the Movement was there Set going.

2. That wheresoever they are Set going, there must be a large and exact Meridian Line prepared by a Perpendicular, set up in an Horizontal Circle, as is Mathematically taught, &c.

3. That an Equal time be spent in Winding it up, as near as possible; for which no time need be allowed, if Equated for 10 or 12 Months; because the time required will be the same for the same time, and greater for a greater, &c.

4. That the said Movement be carefully kept in one and the same Temperature of Air, as near as possible.

5. That after this Movement is fully proved, if it should not (by reason of any sudden change of Air) agree to a Minute, yet it ought not in this respect to be altered or much regarded, because the moistness of one Month, may compensate the Dryness of another: So that by such time as you are arrived to the end of your Voyage, the Movement may even Rectifie or Correct it self.

6. It is to be noted, That if the Sun comes too soon to the Meridian, it will also come too soon to the Horizon; and therefore, Watch too fast, or too slow, must be Added or Subtracted, as aforesaid.

7. That it be always wound up, as near as may be, at one certain time of the Day, (*viz.*) if at Sea, about Twylight, or Break of Day, as may be thought expedient. And if it should happen to be omitted, so long as to stand still; you must not set it going again, but by taking it assunder, and putting the two Indexes together, and bringing the Movement to the Meridian Line, as at first directed. Therefore when at Sea be careful of such Omissions.

8. It is to be noted, That altho' every Minute of Time, does Answer to about 15 or 16 Miles *English*, under the *Equinoctial*; yet it is not above 9 or 10 Miles in our Latitude to one such Minute. And altho' it be so many under

the Equinoctial, yet *there* the Sun's Altitude will be taken very exact, because 90 degrees of Altitude, is for 6 hours of Time; whereas in our Latitude, it is but 62 degrees, for 8 hours of Time; both which Advantages will Facilitate the Practice. I shall at present conclude with acquainting,

That if a common Minute Watch (by Filing or Loading the Ballance, or otherwise) be made to go, or does already go, any unknown quantity of Time, either too Fast, or too Slow; by this Device, we may find the Hour and Minute of the Day by such a Watch, more exactly than can be done by the best timed common Watch whatever. And this even without touching it, other than by Winding up the Spring as usual.

There are other Improvements that may be made of this Discovery, but I reserve them to some other Opportunity.

The FORM of the Book to be kept for each Movement, wherein must be entred the Numbers, &c. (which are to be found at the Returning of the Ship) for finding the Hour and Minute at London.

The Movement, Number 1. was first Set Going <i>Sept. 4. 1714.</i> and Then made one Revolution, in 5 days, 4 hours, and a quarter.							
Degrees of Latitude.	Numbers to be found and used in finding the Hour at <i>London.</i>	Hours spent in producing the said Numbers.	Difference in Numbers.	Seasons in which the Ship was Sailing.	Time when the Movement was set in Motion & Try'd; at & after first Making or Cleaning.	Numbers, shewing how the Motion is retarded by Foulness, on a Tryal of 10 Days at a time.	
I.	II.	III.	IV.	V.	VI.	VII.	Days.
50	29276.2	240		<i>Autumn</i>	<i>Sept. 16. 1714</i>	29276.2	10
52	37962.4	432		<i>Winter</i>	<i>Dec. 4. 1714</i>	29272.5	10
44	37942.	430		<i>Spring</i>	<i>Ma. 20. 1715</i>	29264.	10
48	37912.2	431		<i>Summer</i>	<i>June 1. 1715</i>	29237.4	10
Cleansed <i>Sept. 5. 1715.</i> Then made one Revol. in 5 days, 4 ho. and $\frac{1}{8}$.							
50	29274.2	240		<i>Autumn</i>	<i>Sept. 5. 1715</i>	29274.2	10
52	37957.9	432		<i>Winter</i>	<i>Dec. 17. 1715</i>	29268.	10
44	37942.	430		<i>Spring</i>	<i>Ma. 16. 1715</i>	29264.	10
48	37907.5	431		<i>Summer</i>	<i>June 8. 1716</i>	29232.7	10

Note, That the Numbers to be used after *Cleaning*, (*viz.* after *Sept. 5. 1715.*) are found by Subtracting or Adding the Difference between the two Numbers in the Seventh Column, (*viz.* that of *Sept. 5.*) and that of *Sept. 16.* (which is only 2.) out of the former Numbers to be used for the same Latitude; on which there will remain 29274.2 for the *New Number* to be used in Lat. 50. Deg. in *Autumn*, instead of the former, until the next *Cleaning*, (which must be only *Annually*) and so of the rest, from one *Cleaning* to another. And this method must be used for the respective Latitudes, according as Observations shall be made therein, and Numbers found to answer thereunto.

Note also, That this Experiment may be sufficiently Tryed and Proved, by only two or three Voyages, to the South-West parts of *Ireland*, or to *Newfoundland* at farthest.

But notwithstanding what has been said, It is chiefly to be noted, That altho' I have proposed (for Example sake) that the General Numbers should be found, by Sailing to *Newfoundland*, *Virginia*, *Barbados*, &c. and returning again to a *Meridian Line* at *London*: Yet if the *Honourable Persons* before-mentioned in this Title-Page, to whom the whole is most humbly submitted, will please to appoint a Ship for this purpose, the said Numbers may be much better found in manner following; (that is to say) By ordering that the said Ship shall first make a true *Meridian Line*, at or near the Lat. of 60 Degrees; and thereby set the Movements going, in manner aforesaid: After which let them Sail about 40 or 50 Leagues distant from the Shore, and there Cruise *Northward* and *Southward* for about a Month's time; after which let them return again to the same *Line*, and find the Numbers, and also enter them, as before directed. And having so done, let them make another *Meridian Line*, about 15 Deg. more Southward, and there do in like manner: And so at the like Distance to the *Equinoctial*. After which, let them begin again, and repeat the same Process for three or four times: By which means the Numbers will be found, and the different Motion of the Movement discover'd, for all Latitudes and Seasons, as near as needful.

And being returned home, by these Numbers we may find the true *Difference* of Motion, in and for the several Latitudes and Seasons. Which *Difference* must be entred in the 4th Column of the respective Books, reserved for that purpose. By the help of which said *Differences*, you may make as many Movements as may be needful, at any time; as also Books and Numbers peculiar to each Movement. The *Rules* for which will be imparted to any such Watchmaker, as may Reasonably desire the same.

FINIS.

*** END OF THE PROJECT GUTENBERG EBOOK A NEW
DISCOVERY FOR FINDING THE LONGITUDE ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.